

Memory

System Memory

The system memory is the place where the computer holds current programs and data in use. The term ‘memory’ is ambiguous because it can refer to different parts of the PC: there are many different kinds of memory that a PC uses. When used by itself, ‘memory’ refers to the main system memory, which holds the instructions that the processor executes and the data that those instructions work with.

ROM

Read-Only Memory. One major type of memory used in PCs is called ROM. ROM is a type of memory that normally can only be read, as opposed to RAM, which can be both read and written. There are two main reasons that read-only memory is used for certain functions within the PC: permanence and security. Permanence means that the values stored in ROM are always there, whether the power is on or not. For this reason, it is called non-volatile storage. Security refers to the fact that ROM cannot easily be modified, and thus provides a measure of security against accidental (or malicious) changes to its contents.

The most common example of ROM is the system BIOS program. Having this in a permanent ROM means it is available when the power is turned on so that the PC can use it to boot up the system.

RAM

Random Access Memory. The kind of memory used for holding programs and data being executed is called random access memory. RAM differs from read-only memory (ROM) in that it can be both read and written. It is volatile storage because unlike with ROM, the contents of RAM are lost when the power is turned off. RAM is also sometimes called read-write memory or RWM. RAM is called “random access” because earlier read-write memories were sequential and did not allow random access.

The volatility of RAM means that you risk losing what you are working on unless you save it frequently. RAM is much faster than